Exercise Sheet 27

Use $[X_u, X_v] = X_{u\times v}$ for $\mathfrak{su}(2)$ and coordinates $x_j = \ell(X_j)$ on its real dual. On the sphere $|x| = r > 0$, the KKS form is

$$\omega_x(v, w) = x \cdot (v \times w)/r^2.$$

Prequantisation means

$$F_\nabla = -i\omega/\hbar,$$

with $\hbar > 0$. Exercises 27.1–27.5 are core. Exercise 27.6 is optional.

Exercise 27.1 (The orbit form in local coordinates).

(a) At $x = re_3$, compute ξ_1, ξ_2 and $\omega(\xi_1, \xi_2)$.
(b) On the upper hemisphere use coordinates (x_1, x_2) with $x_3 = (r^2 - x_1^2 - x_2^2)^{1/2}$. Show that $\omega = dx_1 \wedge dx_2/x_3$.
(c) Is the singular coefficient at the equator a singularity of the symplectic form?

Exercise 27.2 (Curvature and a latitude phase). Let $m \in \mathbb{Z}$ and use $A_N = -im(1 - \cos\theta)d\phi/2$.

(a) Compute its curvature and its integral over the whole sphere using the southern chart as needed.
(b) Compute the parallel-transport phase around the latitude $\theta = \theta_0$, oriented by increasing ϕ.
(c) Relate the phase to the symplectic area of the northern cap when $m = 2r/\hbar$.

Exercise 27.3 (Which global group?). For $\hbar = 1$, consider radii $r = 1/4, 1/2, 1, 3/2$.

(a) Which spheres have an ordinary prequantum line bundle?
(b) Which have an equivariant prequantum lift for SU(2), and which for SO(3), with the orbit normalisation?
(c) List the section-space dimensions and spins whenever prequantisation is possible.

Exercise 27.4 (The connection on $O(m)$). Let e_0 be a holomorphic frame on the z chart with $\|e_0\|^2 = (1 + |z|^2)^{-m}$. In this frame set $A^{\mathrm{hol}} = -m\bar{z}\, dz/(1 + |z|^2)$.

(a) Differentiate to obtain $F = m\, dz \wedge d\bar{z}/(1 + |z|^2)^2$.
(b) Pass to the unitary frame $e_N = (1 + |z|^2)^{m/2} e_0$ and compute its connection form.
(c) Set $z = \tan(\theta/2)e^{i\phi}$ and recover A_N from the lecture.

Exercise 27.5 (Classical radius and quantum Casimir). For the spin-$j = m/2$ unitary representation Π, define the Hermitian angular momentum operators $J_i = i\hbar\, d\Pi(X_i)$. They satisfy $\sum_i J_i^2 = \hbar^2 j(j+1)I$. This is the Casimir value from Lie Algebras. No operator-domain question occurs in finite dimension.

(a) Compare this scalar with the squared classical orbit radius r^2, for $r = m\hbar/2$.

(b) For $m = 2$ and $\hbar = 1$, compute both values and the representation dimension.

(c) Use $\chi_m(t_\theta) = \sin((m+1)\theta)/\sin\theta$, valid for $\sin\theta \neq 0$, to explain why the appearance of $m+1$ in a character formula does not change the normalised prequantum period m.

Exercise 27.6 (Optional extension: a half-form choice). The holomorphic cotangent bundle of $\mathbb{CP}^1$ is $\mathcal{O}(-2)$. A square root is $\mathcal{O}(-1)$. Consider replacing the uncorrected section space by $H^0(\mathcal{O}(m) \otimes \mathcal{O}(-1))$.

(a) Use $w = 1/z$ and $dw = -z^{-2}dz$ to check the cotangent-bundle degree.

(b) Compute the new dimension and spin for $m \geq 1$, and the section space for $m = 0$.

(c) Why must this choice be stated explicitly when comparing orbit formulas?